

PROJECT -1

Number Guessing Game

Project Title: Number Guessing Game in Java programming

Project Objective:

The objective of this project is to develop a simple console-based game where the user has to guess a randomly generated number within limited attempts. This project helps in understanding the core concepts of Java Programming such as loops, conditions, exception handling, random number generation, and score calculation.

Project Description:

The Number Guessing Game is an interactive Java application where the computer randomly generates a number between **1 and 100**. The player has to guess the correct number within **7 attempts**. After each guess, the program provides hints:

- **Too High!** → if the guessed number is greater than the generated number
- **Too Low!** → if the guessed number is smaller than the generated number

If the player guesses correctly, a score is calculated based on the number of attempts used.

Technologies Used:

- Java Programming
- Scanner Class (for user input)
- Random Class (for random number generation)

Concepts Used:

- `do-while` loop
- `while` loop
- `if-else` statements
- `try-catch` exception handling
- User input handling
- Random number generation
- Score calculation

Code Explanation:

The program starts by importing the `Scanner` and `Random` classes. The `Scanner` class is used to take input from the user, while the `Random` class generates a random number between 1 and 100. A `do-while` loop allows the user to play the game multiple times. Inside it, a `while` loop continues until the user guesses correctly or the maximum attempts are finished. The `if-else` conditions provide hints to the player. The `try-catch` block handles invalid inputs like letters or symbols. The score is calculated using the formula:

Score Formula:

$$\text{Score} = ((\text{Maximum Attempts} + 1) - \text{Attempts}) \times 10$$

This means:

- Guessing in 1st attempt → 70 points
- Guessing in 2nd attempt → 60 points
- Guessing in 3rd attempt → 50 points
- and so on..

output:

```
Welcome to my 1st project- Number Guessing Game!
Guess a number (Attempts left: 7):
30
Too Low!
Guess a number (Attempts left: 6):
50
Too Low!
Guess a number (Attempts left: 5):
60
Too Low!
Guess a number (Attempts left: 4):
80
Too Low!
Guess a number (Attempts left: 3):
90
Too Low!
Guess a number (Attempts left: 2):
95
Correct! You guessed in 6 attempts.
Your score is: 20/70
Do you want to play again? (y/n)
y
```

```
Guess a number (Attempts left: 7):
30
Too Low!
Guess a number (Attempts left: 6):
50
Too Low!
Guess a number (Attempts left: 5):
60
Too Low!
Guess a number (Attempts left: 4):
70
Too Low!
Guess a number (Attempts left: 3):
80
Too Low!
Guess a number (Attempts left: 2):
90
Too Low!
Guess a number (Attempts left: 1):
95
Too High!
sorry,no attempts and the num is:91
Do you want to play again? (y/n)
n
ok,THANKYOU
```

Advantages:

- Simple and easy to understand
- Beginner-friendly project
- Improves logical thinking
- Demonstrates important Java concepts
- Handles invalid input safely

Conclusion:

This Number Guessing Game is a beginner-level project in Java Programming that demonstrates the practical use of loops, conditions, exception handling, and random number generation. It is a useful project for improving programming skills and understanding game logic development.